

Transformations of quadratics



1

a

x	4	5	6	7	8
y	4	1	0	1	4

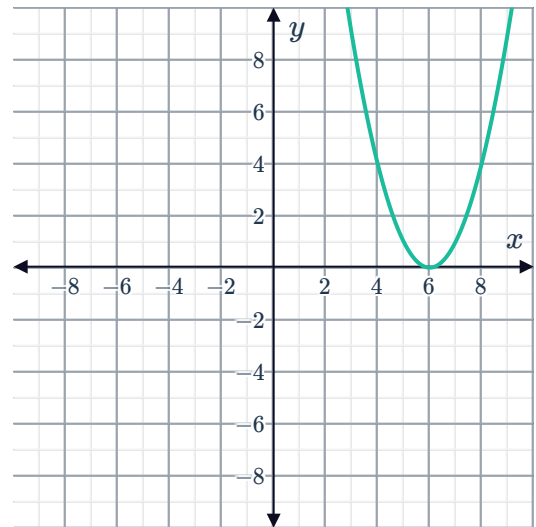
c $x = 6$

e 6 units

b $y = 0$

d $(6, 0)$

f



2

a

x	-8	-7	-6	-5	-4
y	4	1	0	1	4

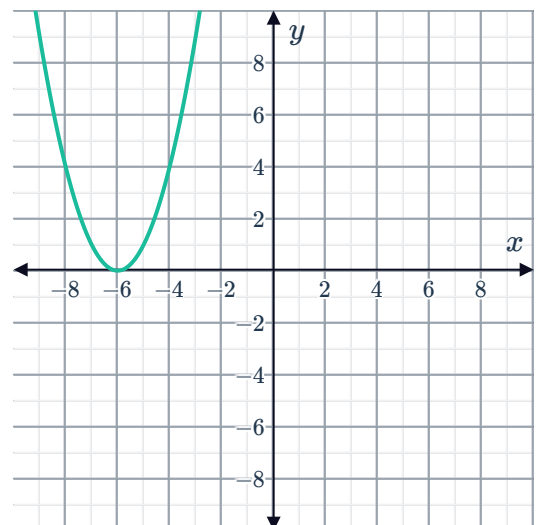
c $y = -6$

e 6 units

b $y = 0$

d $(-6, 0)$

f





3

a $k = -4$

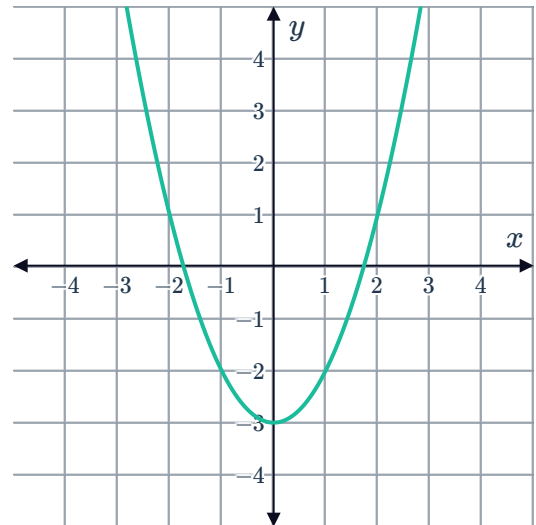
b Horizontal translation of 4 units

4

a

x	-2	-1	0	1	2
y	1	-2	-3	-2	1

b



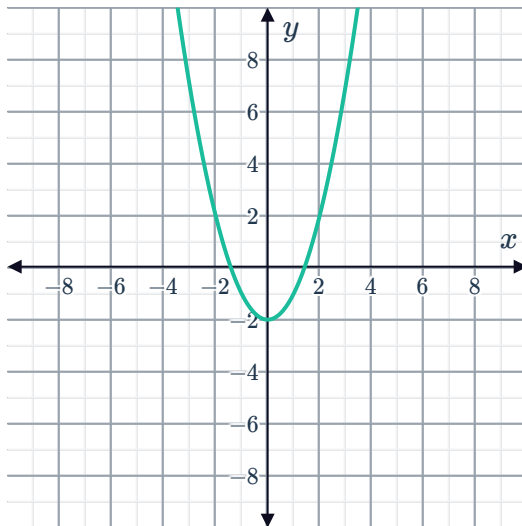
c $y = -3$

d Vertical shift

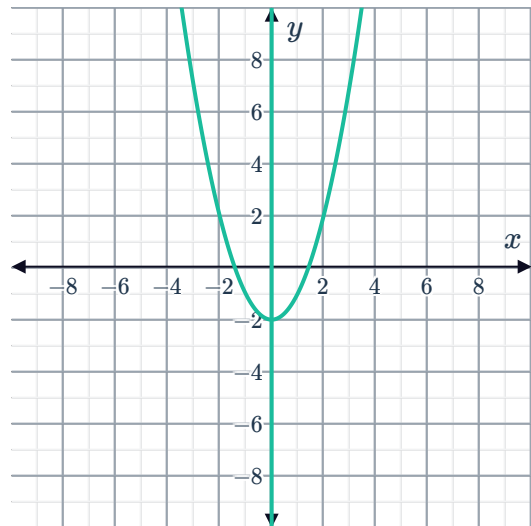
5

a

i

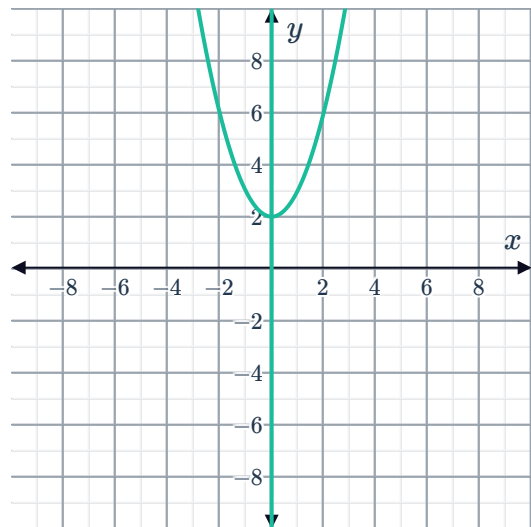
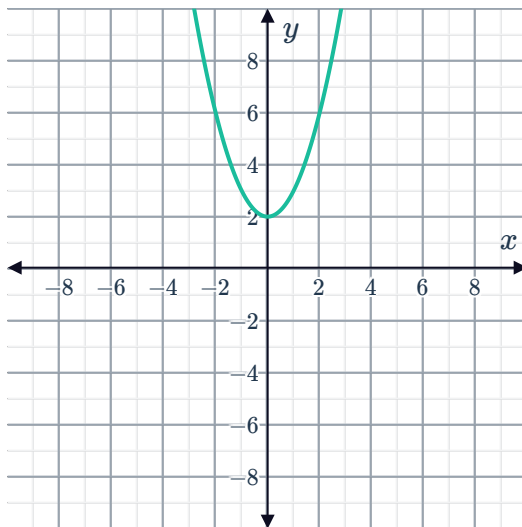


ii



iii $(0, -2)$

b
i



iii (0, 2)

6

a $y = 3$

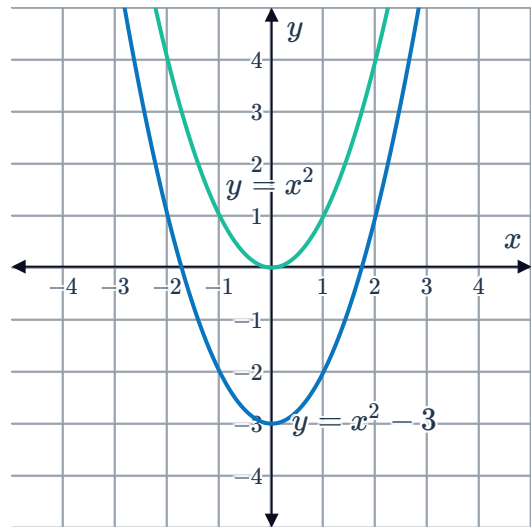
b $y = -4$

c 7 units

7

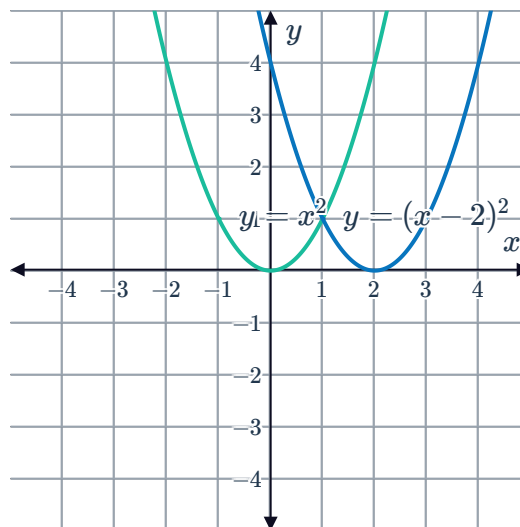
a
i Move the graph downwards by 3 units.

ii



b

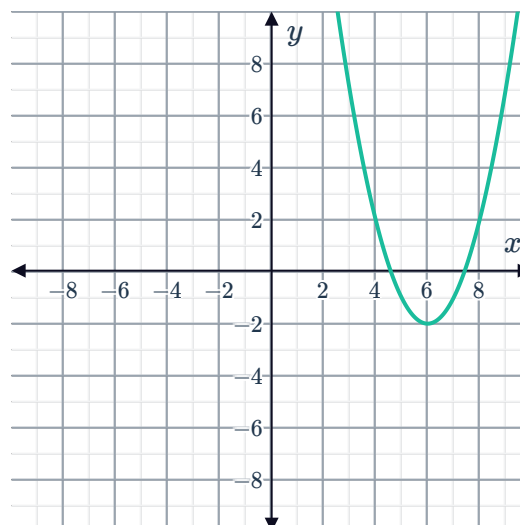
- i Move the graph to the right by 2 units.



8

a $y = (x - 6)^2 - 2$

b



9

a $y = x^2 + 2$

b $y = (x - 3)^2 - 3$

10

a

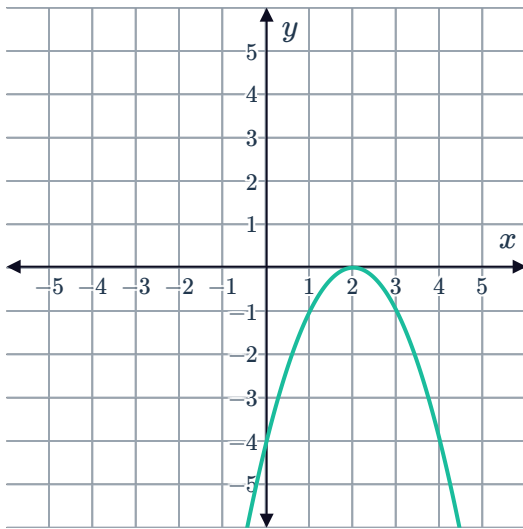
x	0	1	2	3	4
y	-4	-1	0	-1	-4

b $y = 0$

c $x = 2$

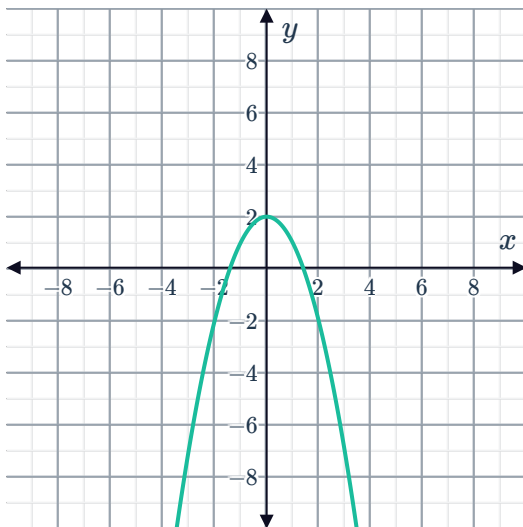
d $(2, 0)$

e



11

a



b (0, 2)

12

a No

b No

c No

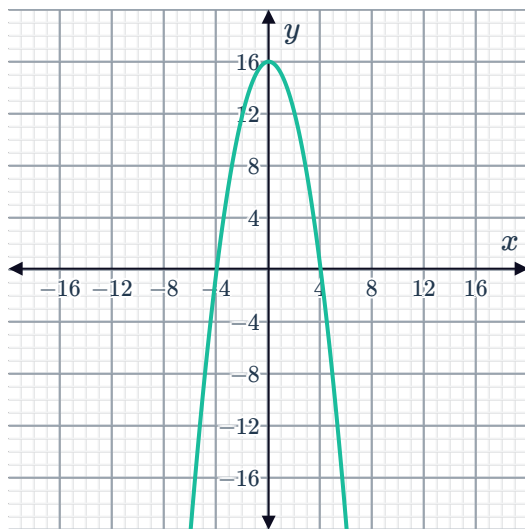
d Yes

13

a $x = \pm 4$

b $y = 16$

c



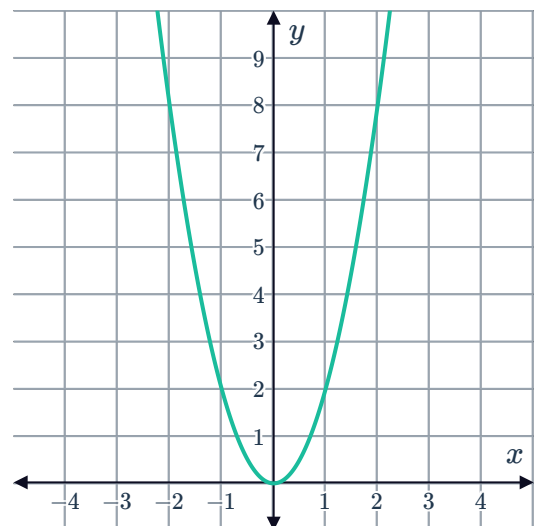
(0, 16)

14

a

x	-2	-1	0	1	2
y	8	2	0	2	8

b



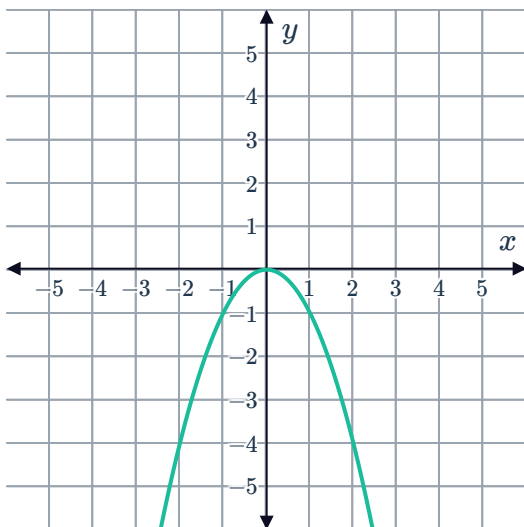
c It narrows the graph.

15

a $a = -1$

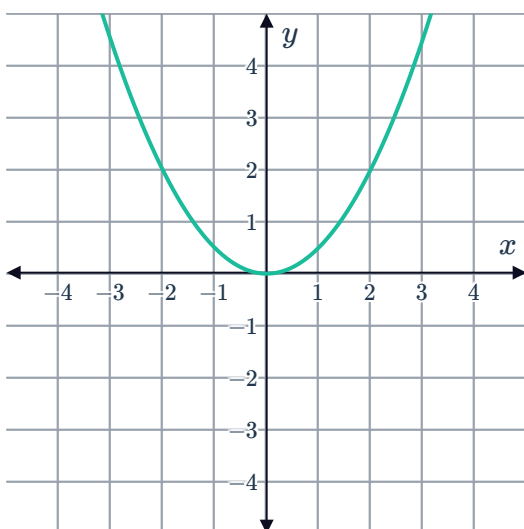
b (0, 0)

c



16

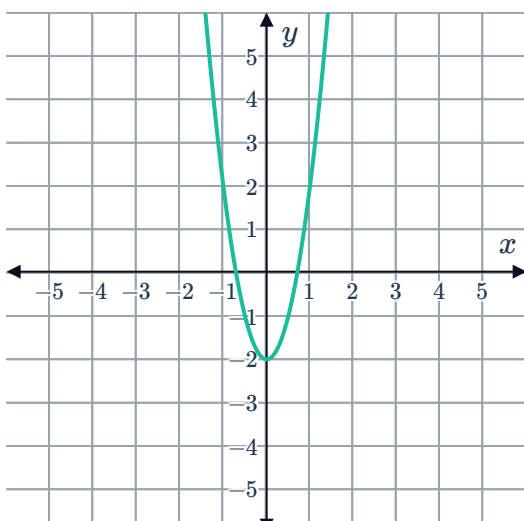
a



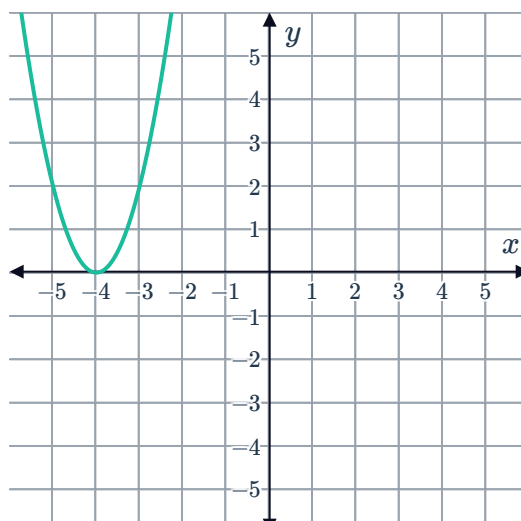
b $y = \frac{1}{2}x^2$

17

a



b





18

a No

b No

c No

19

a 6 units up

b 0 units up or down

c 7 units down

20

a 3 units right

b 7 units left

21

a

i 0 units up or down

ii 0 units left or right

iii No

iv Yes

b

i 2 units down

ii 0 units left or right

iii Yes

iv No

c

i 0 units up or down

ii 4 units right

iii Yes

iv No

d

i 4 units down

ii 3 units right

iii No

iv Yes

22

a $y = 10x^2$

b $y = -5x^2$

c $y = x^2 + 10$

d $y = (x - 2)^2$

e $y = -x^2 - 7$

f $y = 8(x + 4)^2$

g $y = -(x - 6)^2 + 5$

23

a No

b No

c No

d Yes